

MAGIC SCISSORS

Ultra Luxury Unisex Salon & Aesthetic Studio — Technical Requirements Document (TRD)

DOCUMENT VERSION	ENGINE STACK	3D GRAPHICS RIG	CLOUD & DATABASE
2.0.0 (Architecture Baseline)	Vite 5 · ES6+ · TypeScript	Three.js WebGL Engine	Supabase & Netlify CDN

1 ARCHITECTURE TOPOLOGY & MULTI-PAGE BUILD PIPELINE

Magic Scissors is architected around a **decoupled multi-page application (MPA) paradigm** powered by Vite 5 and Rollup. This deliberately avoids Single Page Application (SPA) client-side routing hydration penalties, ensuring near-instant First Contentful Paint (FCP < 0.9s), zero Cumulative Layout Shift (CLS < 0.02), and optimal search engine indexation across every salon portal.

```
Client Browser (Desktop / iOS Safari / Android Chrome)
├─ Netlify Global Edge Anycast CDN (HTTP/2, Brotli/Gzip, Global Edge SSL)
│   └─ Multi-Page HTML Entrypoints (Zero-Hydration Static Endpoints):
│       ├── index.html (Flagship Showcase, 3D Hero, Molten Gallery)
│       ├── about.html (Artisanship, Hygiene Standards, Team)
│       ├── services.html (Treatment Catalog, Pricing Modals)
│       ├── gallery.html (Spatial Bento Grid, Lightbox)
│       ├── franchise.html (Investment Tiers, Capacity Matrix)
│       └── contact.html (Appointments, Dialpad, Google Maps)
│
│   └─ Client Runtime Modules:
│       ├── Three.js WebGL Engine (js/3d/hero-3d.js + device-detection.js)
│       ├── Theatrical Intro State Engine (js/intro.js)
│       ├── Molten Ring 3D Carousel (components/ui/molten-gallery.tsx)
│       └── UI Feedback Engine (js/ui-feedback.js: FormValidator, ButtonLoader, Toasts)
│
│   └─ Backend & Gateway Integrations:
│       ├── Direct WhatsApp Gateway (Sanitized client URI compilation)
│       └── Supabase PostgreSQL Cloud (Appointments table, RLS, real-time channels)
```

2 CODEBASE DIRECTORY STRUCTURE & FILE MAPPING

DIRECTORY / FILE	TECHNOLOGY & SCOPE	ARCHITECTURAL ROLE & DESCRIPTION
index.html ... contact.html	HTML5 Semantic Documents	Discrete multi-page entrypoints; structured with skip links, ARIA landmarks, and high-priority hero preloads.
css/main.css	Vanilla CSS3 Design Tokens	Global CSS variables, typography tokens, fluid clamp functions, safe-area variables, button click ripple effects.
css/components.css	Component Styles	Sticky luxury navigation, brand logo styling, modal backdrops, service cards, footer layout harmony.
css/responsive.css	Adaptive Media Queries	Granular responsive rules for mobile (320px–480px), tablet (768px–1024px), desktop, and ultra-wide 4K viewports.
css/intro.css	Theatrical Keyframe Engine	Heavy velvet curtains, salon chair track animation, amber light flood, and logo blur-to-sharp shimmer.
css/molten-gallery.css	3D Carousel Styling	3D transform perspective, curved glass borders, illuminated champagne glow, gesture touch physics.
js/3d/hero-3d.js	Three.js WebGL Module	Interactive metallic shears viewport, studio lighting rig, dynamic mouse tracking, and mobile gyro damping.
js/3d/device-detection.js	Hardware Benchmarking	Evaluates client GPU, memory, and CPU core count; switches seamlessly between 3D canvas and fallback still photography.

DIRECTORY / FILE	TECHNOLOGY & SCOPE	ARCHITECTURAL ROLE & DESCRIPTION
js/intro.js	Deterministic State Engine	8-state sequence managing chair travel, mechanical scissor click audio, curtain parting, and session storage bypass.
js/ui-feedback.js	Production Feedback Engine	Inline FormValidator, ButtonLoader (Zero-CLS), ToastManager, and StylePhotoUploader dropzone.
components/ui/molten-gallery.tsx	TypeScript React Component	Modular Molten Ring 3D carousel component with touch drag momentum and editorial metadata pill.
scratch/test-suite.js	Node.js Automated Test Harness	74 unit and DOM assertions verifying form validation, loaders, error classifiers, and image decoders.
supabase_schema.sql	PostgreSQL Schema	Defines appointments table, check constraints, Row-Level Security (RLS) policies, and service role privileges.

3 CSS DESIGN SYSTEM & TOKEN ARCHITECTURE

The styling architecture in `css/main.css` establishes an Italian luxury salon aesthetic. All interface elements reference predefined tokens to guarantee absolute visual harmony across every viewport:

```
:root {
  /* Warm Luxury Aesthetic Palette: Warm Alabaster Silk, Champagne Bronze, Deep Noir Charcoal */
  --bg-primary: #FAF8F5;
  --bg-secondary: #F3EFEB;
  --bg-card: rgba(255, 255, 255, 0.96);
  --bg-glass: rgba(255, 255, 255, 0.88);

  /* Terracotta & Champagne Bronze Brand Accent */
  --accent-primary: #B86B49;
  --accent-light: #D49A7E;
  --accent-dark: #8F4A2B;
  --accent-gradient: linear-gradient(135deg, #B86B49 0%, #9E5232 50%, #753418 100%);
  --accent-glow: 0 4px 22px rgba(184, 107, 73, 0.22);

  /* High Contrast Deep Charcoal Noir Typography */
  --text-primary: #15171C;
  --text-secondary: #4A4E58;
  --text-muted: #7A808C;
  --font-serif: 'Cinzel', 'Cormorant Garamond', Georgia, serif;
  --font-display: 'Cinzel', serif;
  --font-editorial: 'Cormorant Garamond', Georgia, serif;
  --font-sans: 'Plus Jakarta Sans', -apple-system, BlinkMacSystemFont, sans-serif;

  /* Safe Area Insets for iOS & Android Notches / Dynamic Island */
  --sat: env(safe-area-inset-top, 0px);
  --sar: env(safe-area-inset-right, 0px);
  --sab: env(safe-area-inset-bottom, 0px);
  --sal: env(safe-area-inset-left, 0px);
}
```

4 THREE.JS WEBGL 3D PIPELINE & LIGHTING MATH

The interactive 3D hero showcase (`js/3d/hero-3d.js`) renders a real-time pair of metallic shears inside a responsive canvas.

- **Perspective Camera:** Field of view 45°, near clipping 0.1, far clipping 1000. Positioned at `(0, 0, 4.2)`.
- **Tone Mapping:** `renderer.toneMapping = THREE.ACESFilmicToneMapping` with exposure 1.15 to prevent blown-out metallic reflections.
- **Studio Lighting Geometry:**
 - **Ambient Fill:** Warm studio bounce (`0xFFFAF0` , intensity 0.65).
 - **Directional Overhead Key:** 45-degree angled key light (`0xFFE8D6` , intensity 1.25) simulating salon downlighting.
 - **Warm Champagne Rim:** Backlight placed behind shear blades (`0xB86B49` , intensity 1.85) creating glowing contour edges.

- **Material Shading:** `MeshStandardMaterial` configured with `metalness: 0.88` and `roughness: 0.12` with an HDR environment reflection map.
- **Damping Physics:** Cursor tracking uses linear interpolation (lerp) with a damping coefficient of `0.05` for smooth cinematic inertia.

5 HARDWARE & DEVICE DETECTION ENGINE (`js/3d/device-detection.js`)

To protect Core Web Vitals on budget smartphones, client capabilities are audited prior to initializing the WebGL context:

```
export function evaluateDeviceTier() {
  const isMobile = /Android|webOS|iPhone|iPad|iPod|BlackBerry|IEMobile|Opera Mini/i.test(navigator.userAgent);
  const memory = navigator.deviceMemory || 4; // Device RAM in GB
  const concurrency = navigator.hardwareConcurrency || 4; // CPU core count
  const connection = navigator.connection;
  const isSlowNetwork = connection && (connection.saveData || /2g|slow-2g/i.test(connection.effectiveType));

  // Constrained hardware conditions
  if (isMobile && (memory < 4 || concurrency < 4 || isSlowNetwork)) {
    return { tier: 'LOW_POWER', use3D: false, dpr: 1 };
  }
  return { tier: 'HIGH_PERFORMANCE', use3D: true, dpr: Math.min(window.devicePixelRatio, 2) };
}
```

6 THEATRICAL INTRO STATE MACHINE & AUDIO ENGINE (`js/intro.js`)

STATE	TRIGGER / TIMING	DOM & ANIMATION EXECUTION	AUDIO & STORAGE EFFECTS
STAGE_INIT	Page Mount	Overlay mounted (<code>z-index: 1000</code>). Curtains closed.	Inspects <code>sessionStorage</code> .
CHAIR_MOTION	+300ms	Salon chair track activates high-velocity travel across stage.	Visual spatial cue.
SCISSOR_SNIP	+1100ms	Mechanical shears perform 3 rapid open-close snips.	Web Audio oscillator metallic click.
CUE_ACTION	+1800ms	<code>#btnGetStarted</code> reveals with animated accent pulse ring.	Interactive cue ready.
CURTAIN_PART	User Tap / Click	Left panel <code>translateX(-100%)</code> , Right panel <code>translateX(100%)</code> .	Cubic-bezier(0.16, 1, 0.3, 1).
LIGHT_FLOOD	Transition Start	Warm amber radial gradient opacity interpolates to 1.0.	Studio backdrop reveal.
LOGO_REVEAL	+400ms	Monogram dissolves from <code>blur(12px)</code> to sharp with gold sweep.	Visual brand imprint.
DISSOLVE	+1400ms	Full overlay fades out; 3D shears canvas initializes.	Sets <code>sessionStorage</code> key.

7 MOLTEN RING 3D CAROUSEL MATHEMATICAL SPECIFICATION

The Molten Ring gallery (`components/ui/molten-gallery.tsx`) arranges N cards on an elliptical 3D cylindrical cylinder. For card index $i \in [0, N-1]$ and rotation offset θ_i :

```
Angle Calculation:      θ_i = (2π / N) × i + θ_rot
Horizontal Coordinate:  X_i = R_x × sin(θ_i)
Depth Coordinate:       Z_i = R_z × (cos(θ_i) - 1)
Dynamic Scale Factor:   Scale_i = ((Z_i + 2 × R_z) / (2 × R_z)) × S_factor
Card Transform Matrix:  transform = translate3d(X_i, 0, Z_i) scale(Scale_i)
```

Inertial Touch Physics: During mobile touch swipe, pointer velocity $v = \Delta x / \Delta t$ is sampled. Upon `touchend` , momentum decay is computed on every animation frame: $v_{t+1} = v_t \times 0.94$. When $|v| < 0.0001$, the carousel

smoothly resumes auto-rotation at \$0.002\text{rad/frame}\$.

8 SUPABASE DATABASE SCHEMA & RLS POLICIES (`supabase_schema.sql`)

```
-- 1. Create Appointments Table with Status Constraints
CREATE TABLE appointments (
  id UUID DEFAULT gen_random_uuid() PRIMARY KEY,
  client_name VARCHAR(120) NOT NULL,
  client_phone VARCHAR(20) NOT NULL,
  service_id VARCHAR(80) NOT NULL,
  preferred_date DATE NOT NULL,
  preferred_time VARCHAR(50) NOT NULL,
  stylist_preference VARCHAR(80),
  notes TEXT,
  status VARCHAR(30) DEFAULT 'PENDING' CHECK (status IN ('PENDING', 'CONFIRMED', 'CANCELLED', 'COMPLETED')),
  created_at TIMESTAMP WITH TIME ZONE DEFAULT NOW()
);

-- 2. Performance Indexing
CREATE INDEX idx_appointments_date ON appointments (preferred_date);
CREATE INDEX idx_appointments_status ON appointments (status);

-- 3. Row-Level Security (RLS) Configuration
ALTER TABLE appointments ENABLE ROW LEVEL SECURITY;

-- Allow anonymous client inserts from booking form
CREATE POLICY "Public insert allowed" ON appointments FOR INSERT WITH CHECK (true);

-- Restrict read & update operations exclusively to authenticated salon receptionist accounts
CREATE POLICY "Staff read allowed" ON appointments FOR SELECT USING (auth.role() = 'authenticated');
CREATE POLICY "Staff update allowed" ON appointments FOR UPDATE USING (auth.role() = 'authenticated');
```

9 DUAL-CHANNEL BOOKING & SANITIZATION PIPELINE

When a user submits the appointment form, the payload traverses a dual validation and dispatch pipeline:

- **Client Sanitization:** Name trimmed (≥ 2 chars); phone numbers stripped of non-digit characters and validated against 10-digit Indian mobile pattern (`/^[6-9]\d{9}$/`); dates validated against `new Date().toISOString()` to prevent past reservations.
- **Channel 1 (Direct WhatsApp API):**

```
const waText = encodeURIComponent(
  `*New Magic Scissors Appointment Request*\n` +
  `Name: ${data.clientName}\nPhone: ${data.clientPhone}\n` +
  `Service: ${data.serviceId}\nDate: ${data.preferredDate}\nTime: ${data.timeSlot}\n` +
  `Notes: ${data.notes || 'None'}`
);
window.open(`https://wa.me/919960135849?text=${waText}`, '_blank');
```
- **Channel 2 (Supabase Asynchronous Logging):** Concurrently dispatches an asynchronous `INSERT` request to Supabase PostgreSQL, ensuring zero client data loss even if the user closes the WhatsApp tab.

10 AUTOMATED VERIFICATION SUITE (`scratch/test-suite.js`)

The test suite executes 74 assertions with zero external testing dependencies in under 2 seconds:

TEST SUBSYSTEM	ASSERTIONS	VERIFIED FUNCTIONAL & DOM BEHAVIORS
ButtonLoader	9 Passed	Zero-CLS dimension preservation, <code>aria-busy="true"</code> , loading spinner injection, state restoration.
ErrorClassifier	8 Passed	Accurate categorization of Network, Auth, Registration, Timeout, and File Upload error boundaries.

TEST SUBSYSTEM	ASSERTIONS	VERIFIED FUNCTIONAL & DOM BEHAVIORS
FormValidator (Appointment)	14 Passed	Name length check, phone regex, mandatory service/date/time, past-date rejection, valid payload output.
FormValidator (Franchise)	2 Passed	Empty state validation rejection, filled franchise submission approval.
Skeletons & EmptyState	7 Passed	Shimmer skeleton HTML generation for service cards, appointment forms, and empty state fallbacks.
ToastManager	5 Passed	Toast rendering, class binding, auto-dismiss timeouts, retry callback execution.
StylePhotoUploader	21 Passed	File dropzone binding, preview rendering, 5 MB file size limit, WebP decoding, object URL revocation.
Gallery & Lightbox Markup	8 Passed	Spatial Bento view array integrity, 6 bento cards, lightbox ID and class synchronization across all HTMLs.

11 CI/CD & DEPLOYMENT PIPELINE

PIPELINE STEP	AUTOMATION TOOL / SCRIPT	RELEASE GATE & OUTPUT
1. Local Verification	<code>workflow.bat</code>	Runs <code>npm install</code> , executes <code>scratch/test-suite.js</code> (must exit 0), validates <code>npm run build</code> .
2. Client Staging Preview	<code>deploy-preview-netlify.bat</code>	Compiles preview bundle and generates live preview URL for client sign-off.
3. Production Release	<code>deploy-netlify.bat</code>	Executes production build, transfers <code>dist/</code> assets to Netlify CDN edge nodes globally.
4. CI Automation	<code>.github/workflows/</code>	Automated GitHub Actions workflow building and deploying previews on repository push.

Magic Scissors — Studio Salons · Technical Requirements Document (TRD) · Version 2.0.0 · October 2026